Linear regression:

$Y \; N\times 1 \quad X \; N\times D \quad w \; D\times 1$

- $\mathcal{L}(w) = \frac{1}{2N}(Y - Xw)^T(Y - Xw)$
- $\nabla \mathcal{L}(w) = -\frac{1}{N}X^T(Y - Xw)$ $O(ND + N + ND + D)$
- $\nabla^2 \mathcal{L}(w) = \frac{1}{N}X^TX$ $O(ND^2)$ — positive semi definite because eigenvalues are positive. Check by SVD
- $w^* = (X^TX)^{-1}X^Ty$ $\underbrace{O(D^3)}_{\text{inversion}} + \underbrace{O(ND^2)}_{X^TX} + \underbrace{O(ND)}_{X^Ty} + \underbrace{O(D^2)}_{(X^TX)^{-1}X^Ty}$ — invertible if $\text{rank}(X) = D$
- High condition number of data matrix $N\times D$ i.e. the columns/features are highly correlated leads to very sensitive $(X^TX)^{-1}$ and huge parameters.

ML estimator $\hat{\theta}$: ① consistent ② $\mathcal{N}(\theta, \frac{1}{N}\{I(\theta)\}^{-1})$ ③ Achieves Cramer-Rao, $Var(\hat{\theta}) \geq \frac{1}{N}\{I(\theta)\}^{-1}$ ④ $\frac{N-1}{N}\hat{\sigma}^2$

True risk for classification: $E_D[\mathbb{1}\{y \neq g(x)\}] = P_D(y \neq g(x))$

Bayes classifier: $\arg\min_{y \in \{0,1\}} P(Y \neq y \mid X = x)$

Ridge:

- $\mathcal{L}_R(w) = \frac{1}{2N}(Y - Xw)^T(Y - Xw) + \lambda\|w\|_2^2$
- $\nabla \mathcal{L}_R(w) = -\frac{1}{N}X^T(Y - Xw) + 2\lambda w$ $O(ND)$
- $w_r^* = (X^TX + 2N\lambda I)^{-1}X^Ty$ $O(d^3 + nd^2)$
- $w_r^* = X^T(XX^T + 2N\lambda I_n)^{-1}y$ $O(n^3 + dn^2)$
- $W_* = X^T\alpha_*$ and $\alpha_* = \arg\min_\alpha \frac{1}{2}\alpha^T(XX^T + \lambda I_n)\alpha - \alpha^Ty$
- $(X^TX + \lambda I) = U[S + \lambda I]U^T$ always invertible
- SGD step should be equal to the normal loss in expectation, thus for a non-averaged loss, multiply by N
- $n\times m \times m\times p$ $O(nmp)$
- SVD of $m\times n$ $O(mn^2 + m^2n)$

Training error: $\frac{1}{|S|}\sum \ell(y_n, f_S(x_n))$

True error: $\mathcal{L}_D(f) = E_D[\ell(y, f(x))]$

Test error: $\frac{1}{|S_{test}|}\sum_{(x,y)\in test} \ell(y_n, f_{S_{train}}(x_n))$

Convexity: $\mathcal{L}(w)$ is convex if

① $f(\lambda x + (1-\lambda)y) \leq \lambda x + (1-\lambda)y$, $\lambda \in [0,1]$

② $\frac{\partial^2 \mathcal{L}(w)}{\partial w \partial w'}$ is positive semi-definite

③ $\mathcal{L}(u) \geq \mathcal{L}(w) + \nabla\mathcal{L}(w)^T(u - w) \quad \forall u, w$

Non-convexity

① Set of discontinuous functions (e.g. set of classifiers because discrete output)

② Proving $f(w_1\pi) = f(w_1, w_2, \pi)$ is not convex.

Construct $p_1 = (w_1', w_2', \pi')$ and p_2 s.t. $f(\frac{1}{2}p_1 + \frac{1}{2}p_2) > \frac{1}{2}f(p_1) + \frac{1}{2}f(p_2)$

Hölder's inequality:

$\sum_{k=1}^n |x_k y_k| \leq \left(\sum_{k=1}^n |x_k|^p\right)^{\frac{1}{p}}\left(\sum_{k=1}^n |y_k|^q\right)^{\frac{1}{q}}$ with $\frac{1}{p} + \frac{1}{q} = 1$

$\leq \|x\|_p \|y\|_q$

① Intersection of convex sets

② $g(x) = f(Ax + b)$

③ $f(x) = \max\{f_1, \dots, f_m\}$ with $\text{dom}(f) = \cap_i \text{dom}(f_i)$

④ $f(x) \geq 0 \; \forall x \Rightarrow g(x) = f^k$

⑤ Positive sum of convex functions

⑥ $f(x) = g(h(x))$ and $g(x), h(x)$ convex and $g(x)$ increasing

⑦ convex + strictly convex = strictly convex

Examples:

- $f(x_1, x_2) = (x_1 - 2x_2)^4 + 2e^{3x_1 + 2x_2 - 5}$
- $f(x) = \log(1 + e^x)$
- $\log\sum_{k=1}^K \exp(w_k^Tx_n)$ w.r.t. W_k (log-sum-exp)

- $g \in \mathbb{R}^D$ is a subgradient of $\mathcal{L}$ at w if $\mathcal{L}(u) \geq \mathcal{L}(w) + g^T(w - u) \; \forall u \in \mathbb{R}^D$

$E[f(x_0) - f_S(x_0)]^2 = Var(\varepsilon) + (E[f_S(x_0)] - f(x_0))^2 + var(f_S(x_0))$

Bound: $P\left(\underbrace{\mathcal{L}_D(f)}_{\text{true risk}} \geq \underbrace{\mathcal{L}_S(f)}_{\text{training risk}} + \sqrt{\frac{(a-b)^2\ln(2/\delta)}{2|S_{test}|}}\right) \leq \delta$

The true risk of the best function with the smallest empirical risk is $+2\sqrt{\frac{(a-b)^2\ln(2K/\delta)}{2|S_{test}|}}$ away from the true risk from the function with the best true risk.

Logistic:

- $\sigma(x) = \frac{e^x}{1 + e^x} = \frac{1}{1 + e^{-x}}$
- $\sigma'(x) = \sigma(x)(1 - \sigma(x))$
- $P(Y = 1 \mid x) = \sigma(x^Tw + w_0)$
- The separating hyperlane is: $w^T = \{v : v^Tw + w_0 = 0\}$
- flat $\xrightarrow{\|w\| \uparrow}$ step function

Changing w_0 shifts the decision region

- $p(y \mid X, w) = \prod_{i=1}^N \sigma(x_i^Tw)^{y_i}(1 - \sigma(x_i^Tw))^{1 - y_i}$
- $\mathcal{L}(w) = \sum_i^N -y_ix_i^Tw + \log(1 + e^{x_i^Tw})$ (convex)
- $\nabla \mathcal{L}(w) = X^T(\sigma(Xw) - y)$ $X \; N\times D \quad y \in \mathbb{R}^n$ $O(ND)$
- $\nabla^2\mathcal{L}(w) = X^TSX \succeq 0$, $S = \text{diag}(\sigma(x_i^Tw)(1 - \sigma(x_i^Tw)))$
- $\mathcal{L}(w) \sim \mathcal{L}(w_t) + \nabla\mathcal{L}(w_t)^T(w - w_t) + \frac{1}{2}(w - w_t)^T\nabla^2L(w_t)(w - w_t)$

$\Rightarrow w_{t+1} = w_t - \gamma_t[\nabla^2\mathcal{L}(w_t)]^{-1}\nabla\mathcal{L}(w_t)$

Representer thm:

For any convex loss ℓ, if $w^* = \arg\min_w \sum_{i=1}^n \ell(x_i^Tw, y_i) + \frac{\lambda}{2}\|w\|^2$

$\exists \alpha_* \in \mathbb{R}^n$ s.t. $W_* = X^T\alpha_*$ (α_* probably computed XX^T)

$\Rightarrow W_* \in \text{span}\{x_1, \dots, x_n\}$

Kernels: $k(x, x') = \phi(x)^T\phi(x)$

- $\alpha, \beta > 0 \quad K = \alpha K_1 + \beta K_2 \Rightarrow \phi = (\sqrt{\alpha}\phi_1, \sqrt{\beta}\phi_2)^T$
- $K = K_1 \cdot K_2 \Rightarrow \phi = \phi_1\phi_2^T$.unravel() (flattened outerproduct)
- $K(x, x') = K(f(x), f(x'))$ • $K(x, x') = f(x)f(x') \; \forall f(x) \in \mathbb{R}$
- $K(x, x') = e^{-(x - x')^2} \Rightarrow \phi(x) = e^{-x^2}(1, \dots, \frac{2^k x^k}{k!}, \dots)$

Mercer's condition: Given kernel k, $\exists \phi$ if

- $\forall x, x'$, $k(x, x') = k(x', x)$
- $\forall n \geq 0 \quad \forall (x_i)_{i=1}^n, K = (K(x_i, x_j))_{i,j=1}^n \succeq 0$

(sufficient, not necessary)

Predicting: We predict $y = \phi(x)^TW_*$ (high dimension)

$\phi(x)^TW_* = \phi(x)^T\phi(X)^T\alpha_* = \sum_{i=1}^n K(x, x_i)\alpha_{*i}$

$\phi(X)^T = \begin{bmatrix}\phi(x_1) \\ \phi(x_n)\end{bmatrix}^T$ $\quad (\underbrace{\phi(X)\phi(X)^T}_{K} + \lambda I_n)^{-1}y$ for kernel ridge

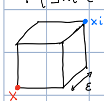
Dimensionality

box = box of size r around the center x_0

$P(x \in \text{box}) = r^d$

Consider n i.i.d points uniform in $[0,1]^d$

$P(\exists x_i \in \text{box}) = 1 - (1 - r^d)^n$



$\|X - x_i\| = \sqrt{\sum_{i=1}^d \varepsilon^2} = \sqrt{d}\,\varepsilon$

$E_{S_{train}}[L(g_{S_{train}})] \leq 2L(g^*) + 4c\sqrt{d}N^{-\frac{1}{d+1}}$

SVM:

Margin: $\{x : |x^Tw| \leq 1\} \Rightarrow$ length of $\frac{2}{\|w\|}$

Hard SVM: $\min \|w\|$ s.t. $\forall i, y_ix_i^Tw \geq 1$

Hinge loss: cost if prediction is incorrect or not confident enough

min le distance to plane for x: $\frac{|w^Tx|}{\|w\|_2}$

Soft SVM:

$\min_w L(w) = \min_w \lambda\|w\|^2 + \sum_{i=1}^n [1 - y_ix_i^Tw]_+$

λ controls the tradeoff between margin size and errors allowed

Minmax:

$\max_\alpha \min_w G(w, \alpha) \leq \min_w \max_\alpha G(w, \alpha)$ — equality if G convex in w, concave in α

$\max_{\alpha \in [0,1]} \alpha x = (x)_+$

$\min_w L(w) = \min_w \max_\alpha G(w, \alpha) = \max_\alpha \min_w G(w, \alpha)$

$= \max_\alpha G(w^*, \alpha)$ with $w^* = \frac{1}{\lambda}\sum_{i=1}^n \alpha_iy_ix_i = \frac{1}{\lambda}X^TY\alpha$

Soft SVM: $\max_{\alpha \in [0,1]^n} \alpha^T1 - \frac{1}{2\lambda}\alpha^T\underbrace{Y\overbrace{XX^T}^{\text{kernel matrix}}Y}_{PSD}\alpha$

$Y = \text{diag}(y)$

$\alpha_i = 0$ outside margin (non support point)

$\alpha_i \in [0,1]$ on the margin (essential support vector)

$\alpha_i = 1$ inside the margin (bound support vector)

Coordinate descent: $\min F(x)$, $x \in \mathbb{R}^D$

- Initialize: $x^0 = (x_1^0, \dots, x_n^0)$
- $x_n^{t+1} = \max_{y \in \mathbb{R}} F(x_1^{t+1}, \dots, y, \dots, x_n^{t+1})$
- $F(x^t) \geq F(x^t) \geq \dots \geq F(x^*)$

$O(D)$ update for SVM

Ethics: A: class Y: predicted label R: score

Independence: $R \perp A$ (equal acceptance rate)

Separation: $R \perp A \mid Y$ (equal false positive and negative rate)

Sufficiency: $A \perp Y \mid R$ ($P(Y=1 \mid R=r, A=a) = P(Y=1 \mid R=r, A=b)$)

Vector calculus

$\frac{\partial}{\partial x_\lambda} x^TAx = \frac{\partial}{\partial x_\lambda}\sum_{i=1}^n\sum_{j=1}^n a_{ij}x_jx_i$

$= \sum_{i=1}^n a_{i\lambda}x_i + \sum_{j=1}^n a_{\lambda j}x_j = (A)_\lambda^Tx + (A)_\lambda x$

- $\boxed{\nabla_x x^TAx = Ax + A^Tx}$
- $\nabla_x b^Tx = b \quad \nabla_x Ax = A$
- $\nabla_W x^TWx = xx^T$

$\begin{bmatrix}a & b \\ c & d\end{bmatrix}\begin{bmatrix}x \\ y\end{bmatrix} = x\begin{bmatrix}a \\ c\end{bmatrix} + y\begin{bmatrix}b \\ d\end{bmatrix}$

$X\begin{bmatrix}| & & | \\ y_1 & \cdots & y_n \\ | & & |\end{bmatrix} = \begin{bmatrix}| & & | \\ Xy_1 & \cdots & Xy_n \\ | & & |\end{bmatrix}$

$UU^T = (u_1 \cdots u_n)\begin{pmatrix}u_1^T \\ \vdots \\ u_n^T\end{pmatrix} = \sum_{i=1}^n u_iu_i^T$

u_i column-vector

Lagrangian: $\max f(x)$ s.t. $g(x) = b$

- $L(x, \lambda) = f(x) + \lambda(b - g(x))$
- Find $x(\lambda) = \arg\min_x L(x, \lambda)$
- Then, find λ^* s.t. $g(x(\lambda^*)) = b$ $\Rightarrow x^* = x(\lambda^*)$

Exp family

- $p(y \mid \eta) = h(y)\exp[\eta^T\overbrace{\phi(y)}^{\text{sufficient statistic}} - \overbrace{A(\eta)}^{\text{cumulant}}]$
- $\int h(y)\exp[\eta^T\phi(y)]dy = e^{A(\eta)}$
- $\eta = g(E[\phi(y)]) \Leftrightarrow E[\phi(y)] = \mu = g^{-1}(\eta)$
- $A(\eta)$ is convex • $\nabla A(\eta) = E[\phi(y)]$
- convex in η • $\nabla^2 A(\eta) = var(\phi(y))$
- $\mathcal{L}(\eta) = -\ln(p(y \mid \eta))$

$= \sum[-\ln(h(y_n)) - \eta^T\phi(y_n) + A(\eta)]$

- $\frac{1}{N}\nabla\mathcal{L}(\eta) = -(\frac{1}{N}\sum_{n=1}^N \phi(y_n)) + E[\phi(y)] = 0$

$\Rightarrow \eta = g(\frac{1}{N}\sum_{n=1}^N \phi(y_n)) = g(\hat{E}[\phi(y)])$

GLM:

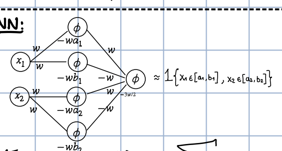
$p(y \mid x, w) = h(y)e^{x^Tw\phi(y) - A(x^Tw)}$

$\nabla\mathcal{L}(w) = X^T[g^{-1}(Xw) - \phi(y)]$ $O(ND)$

Approximation:

- One sigmoid layer: $h(\tilde{\phi}(w(x-a)) - \phi(w(x-b))) \approx h\mathbb{1}\{x \in [a,b]\}$
- Any smooth functions can be $\mathcal{L}^2$ approximated by a sigmoid one-hidden layer NN
- A NN with sigmoid activation and at most two hidden layers can approximate a smooth function in $\mathcal{L}_1$-norm.
- One RELU hidden-layer can give us a point wise approximation of a continuous function

NN:



$\approx \mathbb{1}\{x_1 \in [a_1, b_1], x_2 \in [a_2, b_2]\}$

Filter: $x_{n,m}^{(1)} = \sum_{-2 < k, l < 2} f_{k,l} \cdot x_{n-k, m-l}^{(0)}$

- $f_{k,l}$ is constant • $W * x : \mathbb{R}^D \to \mathbb{R}^D$: Wx (circulant)

Weight sharing: many edges use the same weight

1. Run backprop ignoring weights are shared
2. Sum up the gradients of all edges that share the same weight

Why: Let $f(x, y) : \mathbb{R}^2 \to \mathbb{R}$. Let $g(x) = f(x, x)$. $\frac{\partial g(x)}{\partial x} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}$

Activation functions:

- $\sigma(x) = \frac{1}{1 + e^{-x}}$ smooth, $[0,1]$, vanishing
- $\tanh(x) = 2\sigma(2x) - 1$ smooth, $[-1,1]$, vanishing
- $RELU(x) = (x)_+$ no vanishing gradient, but dying RELU problem
- $L\text{-}RELU(x) = \max(\alpha x, x) \quad 0 < \alpha < 1$

Pooling: down sampling operation

- not overlapping • max or average over the kernel

Batch norm: output $\hat{x}_n^{(\ell)}$

- $\bar{x}_n^{(\ell)} = \frac{x_n^{(\ell)} - \mu_B^{(\ell)}}{\sqrt{(\sigma_B^{(\ell)})^2 + \varepsilon}}$
- $\hat{x}_n^{(\ell)} = \gamma^{(\ell)}\bar{x}_n^{(\ell)} + \beta^{(\ell)}$

Adversarial example from x: $\max_{\hat{x}, \|x - \hat{x}\| \leq \varepsilon} \mathbb{1}\{f(\hat{x}) \neq y\}$

$\rightarrow \max_{\hat{x}, \|x - \hat{x}\| \leq \varepsilon} \tilde{\ell}(\hat{x})$ where $\tilde{\ell}(\hat{x}) = \ell(y g(\hat{x}))$ and $g(x) \in \mathbb{R}$ is NN output

$\rightarrow \max_{\hat{x}, \|x - \hat{x}\| \leq \varepsilon} \tilde{\ell}(\hat{x}) \approx \underbrace{\tilde{\ell}(x)}_{\text{not relevant}} + \max_{\|\delta\| \leq \varepsilon} \nabla_x\tilde{\ell}(x)^T\delta$ where $\delta = x - \hat{x}$

$\rightarrow$ By def: $\hat{x} = x + \delta^*$. To find δ^*, we know $\nabla_x\tilde{\ell}(x) = y\nabla_x g(x)\underbrace{\ell'(y g(x))}_{\leq 0 \text{ (def of loss)}}$

$\rightarrow \ell_2$ norm: $\delta_2^* = \varepsilon \cdot \frac{\nabla_x\tilde{\ell}(x)}{\|\nabla_x\tilde{\ell}(x)\|_2} = -\varepsilon y \cdot \frac{\nabla_x g(x)}{\|\nabla_x g(x)\|_2}$

$\rightarrow \ell_\infty$ norm: $\delta_\infty^* = \varepsilon \cdot \text{sign}(\nabla_x\tilde{\ell}(x)) = -\varepsilon y \cdot \text{sign}(\nabla_x g(x))$

$\rightarrow$ Can iterate and use $\hat{x} = x + \delta^*$ as a new starting point to compute δ^*. Just project back δ^* into $\|\delta\| < \varepsilon$ at each step

Standard training: $\min_\theta R(f_\theta) = E_D[\mathbb{1}f(\hat{x}) \neq y]$

Adversarial training: $\min_\theta R_\varepsilon(f_\theta) = E_D[\max_{\hat{x}, \|x - \hat{x}\| \leq \varepsilon} \mathbb{1}\{f(\hat{x}) \neq y\}]$

$\rightarrow$ relax and use $\min_\theta \frac{1}{N}\sum_{i=1}^N \max_{\hat{x}_i, \|x_i - \hat{x}_i\| \leq \varepsilon} \ell(y g_\theta(\hat{x}_i))$

1. For each x_i, find $\hat{x}_i$ via above
2. Do a gradient step w.r.t. θ using $\frac{1}{N}\sum_{i=1}^N \nabla_\theta \ell(y g_\theta(\hat{x}_i))$ using backprop

Effect of norm:

ℓ_∞: perturbate all coordinates by a small amount

ℓ_{1-0}: changing few coordinates by a lot

Weight decay: $\min L + \frac{\mu}{2}\sum_{\ell=1}^{L+1}\|W^{(\ell)}\|_F^2$ • $(w_{i,j}^{(\ell)})_{t+1} = (1 - \gamma\mu)(w_{i,j}^{(\ell)})_t - \gamma\nabla$

K-means clustering:

$$\min_{z,\mu} \mathcal{L}(z,\mu) = \sum_{n=1}^{N}\sum_{k=1}^{K} z_{nk} \| x_n - \mu_k \|_2^2$$

s.t. $\mu_k \in \mathbb{R}^D$, $z_{nk} \in \{0,1\}$, $\sum_{k=1}^{K} z_{nk} = 1$

0. Initialize μ $O(N \cdot K \cdot D)$
1. $z_{nk} = \mathbb{1}\{k = \text{argmin}_j \|x_n - \mu_j\|_2\}$
2. $\mu_k = \dfrac{\sum_{n=1}^{N} z_{nk} x_n}{\sum_{n=1}^{N} z_{nk}}$

- $W^*_{MAP} = \text{argmax } p(y \mid X, w)\, p(w)$
- if $y \in \mathbb{R}^+$, use RELU for output
- Sum of Gaussians is not convex
- Identifiable iff $\theta_1 = \theta_2 \Rightarrow \mathcal{L}(\theta_1) = \mathcal{L}(\theta_2)$
- our SVM assumes $y \in \{-1, 1\}$
- An MLP can never be strictly convex because permutation of any hidden layer nodes would lead to the same minimum (K!)

GMM: $p(X \mid \mu, \Sigma, z) = \prod_{n=1}^{N}\prod_{k=1}^{K} [\mathcal{N}(x_n \mid \mu_k, \Sigma_k)]^{z_{nk}}$

$$p(X, z \mid \mu, \Sigma, \pi) = \prod_{n=1}^{N} p(x_n \mid z_n, \mu, \Sigma)\, p(z_n \mid \pi)$$

$$= \prod_{n=1}^{N}\prod_{k=1}^{K} [\mathcal{N}(x_n \mid \mu_k, \Sigma_k)]^{z_{nk}} \prod_{k=1}^{K} [\pi_k]^{z_{nk}}$$

parameters to learn in GMM:
- $\mu \in \mathbb{R}^{D \cdot K}$
- $\Sigma \in \mathbb{R}^{D^2 \cdot K}$
- $\pi \in \mathbb{R}^K$

$\theta = \{\mu, \Sigma, \pi\}$

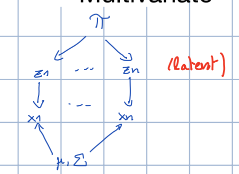
Marginal likelihood: $p(x_n \mid \theta) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x_n \mid \mu_k, \Sigma_k)$

$$f_{\mu,\Sigma}(x) = \frac{1}{(2\pi)^{N/2}|\Sigma|^{1/2}} \exp\left[-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\right]$$

Thanks to this, # of parameters is $O(D^2 K)$ instead of $O(N)$.

- $p(x_n) = \sum_z p(x, z) = \sum_z p(x \mid z)\, p(z)$

Multivariate



ML: $\min_\theta -\sum_{n=1}^{N} \log\left[\sum_{k=1}^{K} \pi_k \mathcal{N}(x_n \mid \mu_k, \Sigma_k)\right]$

① non-convex ② non-unique optima (switch assignments)
③ not bounded if $\|\Sigma\| \to 0$

EM algorithm:

1. E-step: Compute a lower bound to the cost s.t. it is tight at previous $\theta^{(t)}$
   - $\mathcal{L}(\theta) \geq \underline{\mathcal{L}}(\theta, \theta^{(t)})$
   - $\mathcal{L}(\theta^{(t)}) = \underline{\mathcal{L}}(\theta^{(t)}, \theta^{(t)})$ → defines the value of $\theta^{(t)}$
2. M-step: Update θ:
   $\theta^{(t+1)} = \arg\max_\theta \underline{\mathcal{L}}(\theta, \theta^{(t)})$

$\underline{\mathcal{L}}(\theta, \theta^{(t)})$ is a surrogate to the cost function defined in $\theta^{(t)}$, hopefully convex

EM for GMM:

1. E-step:
   - $q_{kn}^{(t)} = \dfrac{\pi_k^{(t)} \mathcal{N}(x_n \mid \mu_k^{(t)}, \Sigma_k^{(t)})}{\sum_{k=1}^{K} \pi_k^{(t)} \mathcal{N}(x_n \mid \mu_k^{(t)}, \Sigma_k^{(t)})}$
2. Compute new surrogate bound
   - $\mathcal{L}(\theta^{(t)}) = \underline{\mathcal{L}}(\theta^{(t)}, \theta^{(t)}) = \sum_{k=1}^{K} q_{kn}^{(t)} \log \dfrac{\pi_k^{(t)} \mathcal{N}(x_n \mid \mu_k^{(t)}, \Sigma_k^{(t)})}{q_{kn}^{(t)}}$
3. M-step: derive first-order condition from surrogate.
   - $\mu_k^{(t+1)} := \dfrac{\sum_n q_{kn}^{(t)} x_n}{\sum_n q_{kn}^{(t)}}$
   - $\Sigma_k^{(t+1)} := \dfrac{\sum_n q_{kn}^{(t)} (x_n - \mu_k^{(t+1)})(x_n - \mu_k^{(t+1)})^T}{\sum_n q_{kn}^{(t)}}$
   - $\pi_k^{(t+1)} := \frac{1}{N} \sum_n q_{kn}^{(t)}$
   - if $\Sigma_k = \sigma^2 I$ as $\sigma^2 \to 0$, GMM = K-means

Surrogate for GMM:

Given $\sum q_k = 1$, $q_k > 0$ and $r_k > 0$,

$$\log\left(\sum_{k=1}^{K} q_k r_k\right) \geq \sum_{k=1}^{K} q_k \log(r_k)$$

Expectation step:

$$\log \sum_{k=1}^{K} \pi_k \mathcal{N}(x_n \mid \mu_k, \Sigma_k) \geq \sum_{k=1}^{K} q_{kn}^{(t)} \log \frac{\pi_k^{(t)} \mathcal{N}(x_n \mid \mu_k^{(t)}, \Sigma_k^{(t)})}{q_{kn}^{(t)}}$$

with equality when

$$q_{kn}^{(t)} = \frac{\pi_k^{(t)} \mathcal{N}(x_n \mid \mu_k^{(t)}, \Sigma_k^{(t)})}{\sum_{k=1}^{K} \pi_k^{(t)} \mathcal{N}(x_n \mid \mu_k^{(t)}, \Sigma_k^{(t)})} = p(z_{nk} = 1 \mid x_n, \theta^{(t)})$$

Posterior distribution

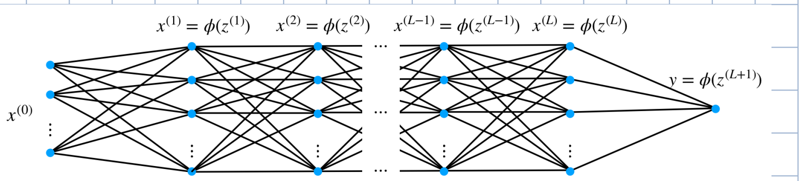
$$p(x_n, z_n \mid \theta) = p(x_n \mid z_n, \theta)\, p(z_n \mid \theta) = p(z_n \mid x_n, \theta)\, p(x_n \mid \theta) = q_n^{(t)}\, p(x_n \mid \theta)$$

EM in general:

Given $p(x_n, z_n \mid \theta)$, the marginal likelihood can be lower-bounded.

$$\theta^{(t+1)} := \arg\max_\theta \sum_{n=1}^{N} E_{z \mid x, \theta}[\log p(x_n, z_n \mid \theta)]$$

generic way to get a surrogate because expectation satisfies Jensen's inequality and we average out the latent variables.



$x^{(l)} \in \mathbb{R}^K$ is row vector of values of nodes at layer l

Forward pass:

$x^{(0)} = x_n \in \mathbb{R}^D$

$z^{(l)} = (W^{(l)})^T x^{(l-1)} + b^{(l)}$

$x^{(l)} = \phi(z^{(l)})$

$L_n = \frac{1}{2}(y_n - f^{(L+1)} \circ \cdots \circ f^{(2)} \circ f^{(1)}(x_n))^2 = \frac{1}{2}(y_n - \tilde{\phi}(z^{(L+1)}))^2$

with $\tilde{\phi}(z^{(L+1)}) = (w^{(L+1)})^T z^{(L+1)}$ or $\text{sign}((w^{(L+1)})^T z^{(L+1)})$

Backward pass:

$$\delta_j^{(l)} = \frac{\partial L_n}{\partial z_j^{(l)}}$$

$$\delta^{(L+1)} = (\tilde{\phi}(z^{(L+1)}) - y_n)\, \tilde{\phi}'(z^{(L+1)})$$

here, square loss but loss function only changes $\delta^{(L+1)}$

$$\delta^{(l)} = (W^{(l+1)} \delta^{(l+1)}) \odot \phi'(z^{(l)})$$

Diff Nash Equ: (θ^*, φ^*) is a diff. Nash Equ. iff.

$\|\nabla_\theta \mathcal{L}(\theta^*, \varphi^*)\| = \|\nabla_\varphi \mathcal{L}(\theta^*, \varphi^*)\| = 0$

$\nabla_\theta^2 \mathcal{L}(\theta^*, \varphi^*) \succ 0 \quad \nabla_\varphi^2 \mathcal{L}(\theta^*, \varphi^*) \prec 0$

θ wants to minimize
φ wants to maximize

GAN:

$$\min_G \max_D \; E_{x \sim p_d}[\log(\underbrace{D(x)}_{\text{prob. of distinguishing well}})] + E_{z \sim p_z}[\log(\underbrace{1 - D(G(z))}_{\text{prob. of being detected}})]$$

The optimum is reached when $p(\text{real data}) = p(\text{generated})$ i.e. $p_g = p_d$

$x \sim p_d$ real sample $[y=1]$ → $D: x \mapsto y \in \{0,1\}$

$z \sim p_z$ → $G: z \to x$ → $G(z)$ fake sample $x \sim p_g$ $[y=0]$

Distances:

KL: $D_{KL}(p_d \| p_g) = E_D\left[\log\left(\frac{p_d(x)}{p_g(x)}\right)\right]$

JS: $D_{JS}(p \| q) = \frac{1}{2} D_{KL}(p \| \frac{p+q}{2}) + \frac{1}{2} D_{KL}(q \| \frac{p+q}{2})$

Distances are not well defined if the supports are not the same

Wasserstein-1 distance:

$$D_W(p_d, p_g) = \inf_{\gamma \sim \Pi(p_d, p_g)} E_{(x,y) \sim \gamma}[\|x - y\|]$$

SVD

$s_i \geq 0$ by convention

$\underset{D \times N}{X} = \underset{D \times D}{U}\, \underset{D \times N}{S}\, \underset{N \times N}{V^T}$

Best rank-K approximation:

$S^{(K)} = \begin{bmatrix} S_K & 0 \\ 0 & 0 \end{bmatrix}$, $U_K = D \times K$, $U_K^T U_K = I_K$, $U_K^T U = [I_K \mid 0]$

$$\tilde{X} = U S^{(K)} V^T = U_K U_K^T X = \sum_{i=1}^{K} s_i u_i v_i^T = \underbrace{U_K}_{W} \underbrace{S^{(K)} V^T}_{Z^T}$$

$$\|X - \tilde{X}\|_F^2 = \sum_{i=k+1}^{D} s_i^2$$

$S^T S = $ diagonal

Properties:

- Columns of V are the eigenvectors of $X^T X$. The non-zero singular values are the square roots of the non-zero eigenvalues of $X^T X$.

$S S^T = $ diagonal

Constructing U from V:
$X^T X = V S^T S V \Rightarrow X^T X v_j = V S^T S V^T v_j = s_j^2 v_j$
$\Rightarrow X X^T (X v_j) = s_j^2 (X v_j) \Rightarrow u_j = X v_j$

$X^T X v_j = s_j^2 v_j$, $X X^T u_j = s_j^2 u_j$, $X v_j = u_j$

PCA

Principal components: $U_K^T X$ $(K \times N)$
Principal directions: columns of U

Assume centered features of matrix
$X X^T$ = covariance matrix $= U S_0^2 U^T$
Let $\tilde{X} = \underbrace{U_K^T X}_{PCA} \Rightarrow \tilde{X}\tilde{X}^T = S_K^2$ (uncorrelated)

Matrix factorization

Two losses:

$$\mathcal{L}(W, Z) = \|X - WZ^T\|_F^2 + \frac{\lambda_w}{2}\|W\|_F^2 + \frac{\lambda_z}{2}\|Z\|_F^2$$

$$\mathcal{L}(w, z) = \sum_{(d,n) \in \Omega} [x_{dn} - (WZ^T)_{dn}]^2 + \frac{\lambda_w}{2}\|W\|_F^2 + \frac{\lambda_z}{2}\|Z\|_F^2$$

① not convex; Example: $\mathcal{L}(w, z) = w \cdot z$ is a saddle function, thus not always above its linearization.
② no unique optima; can change signs or scalar and inverse

$(WZ^T)_{dn} = w_d^T z_n$

ALS updates (no missing):
$Z^T := (W^T W + \lambda_z I_K)^{-1} W^T X$
$W^T := (Z^T Z + \lambda_w I_K)^{-1} Z^T X^T$
$O(k^3 + kDN)$

ALS updates (missing):
$z_n = \left(\sum_{(d,n) \in \Omega} [w_d w_d^T] + \lambda_z I_K\right)^{-1} \sum_{(d,n) \in \Omega} x_{dn} w_d$ for $n = 1, \ldots, N$
$O(k^3 + n_n k^2)$ $n_n = |(d,n) \in \Omega|$
$w_m = \left(\sum_{(m,i) \in \Omega} [z_i z_i^T] + \lambda_w I_K\right)^{-1} \sum_{(m,i) \in \Omega} x_{mi} z_i$ for $m = 1, \ldots, D$
$O(k^3 + n_m k^2)$ $n_m = |(m, n) \in \Omega|$

SGD step:
$f_{dn}(W, Z) = [x_{dn} - (WZ^T)_{dn}]^2 = [x_{dn} - \sum_{i=1}^{K} w_{di} z_{ni}]^2$

$\frac{\partial}{\partial w_{d'k}} f_{dn}(W, Z) = -\mathbb{1}[d' = d][x_{dn} - (WZ^T)_{dn}] z_{nk}$ $O(K)$

$\frac{\partial}{\partial z_{n'k}} f_{dn}(W, Z) = -\mathbb{1}[n' = n][x_{dn} - (WZ^T)_{dn}] w_{dk}$

$\underset{D \times N}{X} = \underset{D \times K}{W}\, \underset{K \times N}{Z^T}$

w_i (word vector or movie vector)
z_j (context vector or user vector)

Co-occurence matrix: n_{ij} = # of contexts where word w_i occurs together with word w_j

If symmetric ⇒ $W^T = Z^T$

For Glove: context = window ∈ {5, 10} and use $\log(n_{ij})$

Word2Vec:
For each word vector w_d,
$\max_{w_d} \sum_{\text{real word w/ context}} \phi(w_d^T w_n) + \sum_{\text{random word w/c}} \phi(w_d^T w_n')$

FastText: $\min_{W,Z} \sum_{s_n \text{ a sentence}} \ell(y_n W Z^T \underbrace{\text{BOW}(s_n)}_{\text{sentence representation}})$ ← (can stream through text)

- FastText learns word vectors and sentence representations which are specific to a supervised classification task.
- The skip-gram model learns a binary classifier for each word
- Language models can predict the next word (multi-class) or predict if the next word is real or fake (binary classifier)

Graph:

$u_i^T x_n$

$u_i^T x_n$

Derivatives:

Using that $z_m^{(l)} = \sum_{k=1}^{K} w_{k,m}^{(l)} x_k^{(l-1)} + b_m^{(l)}$

$$\frac{\partial L_n}{\partial w_{i,j}^{(l)}} = \sum_{k=1}^{K} \frac{\partial L_n}{\partial z_k^{(l)}} \frac{\partial z_k^{(l)}}{\partial w_{i,j}^{(l)}} = \frac{\partial L_n}{\partial z_j^{(l)}} \frac{\partial z_j^{(l)}}{\partial w_{ij}^{(l)}} = \delta_j^{(l)} \cdot x_i^{(l-1)}$$

$$\frac{\partial L_n}{\partial b_j^{(l)}} = \delta_j^{(l)}$$

$$\nabla_{W^{(l)}} L_n = \delta^{(l)} (x^{(l-1)})^T$$